



Department of Computer Science and Engineering
BRAC University
CSE423 (Computer Graphics), Quiz: 1
Set: B, Full Marks: 20, Time: 30 minutes
Thursday 27th February, 2025

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$$\frac{16}{9} = \frac{x}{1080}$$

Student Number:

Name:

Signature of Student:

There are 3 questions in this script in 2 pages. Please make sure that you got all of them.
Answer in the space below each question. Extra papers shall be neither supplied nor are you allowed to use any.

1. Consider a 20 inch laptop screen with 1080p display, and aspect ratio 16:9

(3+3)

- a. Calculate the PPI or Pixel Density

$$\frac{16}{9} = \frac{x}{1080} \Rightarrow x = 1920$$

$$PPI = \frac{\sqrt{(Width)^2 + (Height)^2}}{ScreenSize} = \frac{\sqrt{(1920)^2 + (1080)^2}}{20} = 110.19 \approx 110 \text{ ppi (Ans.)}$$

- b. If the bit-depth is 16, calculate the size (in MB) to store an image with the same number of pixels as the mentioned display.

$$\begin{aligned} \text{Image size (in Bytes)} &= \frac{1920 \times 1080 \times 16}{8} \\ &= 4147200 \text{ Bytes} = 4.1472 \text{ MB} \\ &\approx 4.15 \text{ MB (Ans.)} \end{aligned}$$

2. Answer the following in one or two sentences each:

(2+2)

- a. What is the primary advantage of the DDA algorithm over a simple approach that manually calculates y values for each x value?

The primary advantage of DDA Algorithm is that we can get rid of the "mx" multiplication everytime which decreases our computational cost.

$$y_i = \lfloor mx_i + c \rfloor$$

→ get rid of this multiplication in DDA

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- b. What is a drawback of the DDA algorithm that is solved by the Midpoint Line Drawing Algorithm?

The floating point is not solved in DDA Algorithm as we can still get a floating number from the slope "m".

This floating point problem is solved by the Midpoint Line Algorithm as we are adding discrete values everytime.

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3. Use Midpoint Line Drawing Algorithm to find the first 6 points of the line segment starting at (2, 7) and ending at (35, 17). Show your calculations, preferably in a table as shown in the lectures.

(10)

$$\begin{aligned} dy &= 17 - 7 \\ &= 10 \\ dx &= 35 - 2 \\ &= 33 \end{aligned}$$

$$\begin{aligned} NE &= 2dy - 2dx \\ &= 20 - 66 \\ &= -46 \end{aligned}$$

$$\begin{aligned} E &= 2dy \\ &= 20 \end{aligned}$$

$$\begin{aligned} d_{init} &= 2dy - dx \\ &= 20 - 33 \\ &= -13 \end{aligned}$$

x	y	d	NE (+1, +1) E (+1, 0)	d _{new}	Pix
2	7	-13	E	7	(2, 7)
3	7	7	NE	-39	(3, 7)
4	8	-39	E	-19	(4, 8)
5	8	-19	E	1	(5, 8)
6	8	1	NE	-45	(6, 8)
7	9	-45	E	-25	(7, 9)
8	9	-25			